

CM-001-M1-VS03A

Business Entity Management

Enterprise Design & Runtime Specification

1. Executive Summary

VS03A introduces the Business Entity Registry, the foundation of the metadata-driven Catrack Application Platform (CAP). Administrators can define business entities without writing code while remaining fully integrated with Platform Modules, Runtime, Authorization and Audit.

2. Objectives

Provide production-ready CRUD for Business Entity definitions. The slice manages entity metadata only and intentionally excludes Field Designer, View Designer, Runtime Records and Publishing.

3. Scope

Includes Entity Registry, CRUD, validation, search, filtering, duplicate, archive/restore, audit, authorization, repository, services, APIs, server actions and React Query integration.

4. Architecture

BusinessEntity belongs to a PlatformModule and is consumed by the Runtime Engine. Repository -> Service -> Route Handlers/Server Actions -> React Query -> AG Grid UI. All persistence uses Prisma with PostgreSQL.

5. Entity Metadata

Identity, Runtime, Navigation, Behavior, Features and API metadata are configurable. Entity definitions are metadata-driven and become runtime assets after VS03B publishing.

6. User Interface

Entity List implemented with AG Grid Enterprise supporting search, filtering, sorting, pagination, export and bulk operations. Entity Details uses tabbed pages: General, Navigation, Runtime, API, Advanced and Audit.

7. Validation Matrix

Validate required fields, uniqueness of Entity Code, Entity Name, API Name and Route, reserved keywords, metadata locks and system entity protection. Client and server validation must remain synchronized.

8. Repository & Services

Repository supports CRUD, archive, restore, duplicate and existence checks. Services encapsulate business rules, validation and duplication logic. Transactions protect multi-step updates.

9. API Specification

REST endpoints for list, detail, create, update, delete, archive, restore and duplicate. Server Actions mirror the same operations for Next.js App Router.

10. Security

Authorization integrates with the CAP Authorization Engine. Permissions include View, Create, Edit, Delete, Archive, Restore, Duplicate and Export. Unauthorized operations are hidden and rejected server-side.

11. Audit

All changes generate audit entries capturing user, timestamp, previous values, new values and changed fields.

12. Performance

Server-side paging, filtering and sorting, optimized Prisma queries, avoidance of N+1 queries and efficient cache invalidation through React Query.

13. Testing

Repository, service, validation and API tests covering successful execution, authorization failures and validation failures.

14. Acceptance Criteria

Production-ready implementation with no placeholder logic, TODOs or mock code. Lint, build and Prisma validation must succeed.

15. Implementation Checklist

Integrate Repository, Services, Route Handlers, Server Actions, React Query, Authorization and Audit. Return files modified, files created, database changes, tests added and remaining production risks after implementation.

Appendix A - Anti Gravity Delivery Requirements

Every implementation must end by running `npm run lint`, `npm run build` and `prisma validate`. Fix all issues before returning results. Deliver production-ready code only.